

Database Systems Lab

Lab Task



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24P-0554

Section 4C

Operation 01

.mode table

SELECT * FROM "developers"

```
sqlite> .mode table
sqlite> SELECT * FROM "DEVELOPERS";
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
2	India	5	Java	SQL	Oracle	Spring	18000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
5	Pakistan	1	C				3000
6	UK	12	Python	R	PostgreSQL	Flask	85000
7	Canada	4	JavaScript	Python	MongoDB	Vue	60000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
9	USA	15	C++	Python	SQLite	Qt	110000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
11	France	8	Python	SQL	PostgreSQL	Django	65000
12	Australia	9	JavaScript		MongoDB	React	75000
13	Japan	11	Java	Python	Oracle	Spring	82000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
15	Netherlands	6	Go	Python	PostgreSQL		68000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000
19	India	2	JavaScript	HTML	Firebase	Angular	10000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 02

SELECT "country" , "primary_language" ,
"secondary_language" FROM "developers"

```
sqlite> SELECT "country", "primary_language" , "secondary_language" FROM "developers";
```

country	primary_language	secondary_language
Pakistan	Python	JavaScript
India	Java	SQL
USA	JavaScript	TypeScript
Germany	C#	SQL
Pakistan	C	
UK	Python	R
Canada	JavaScript	Python
Pakistan	Java	Kotlin
USA	C++	Python
India	PHP	JavaScript
France	Python	SQL
Australia	JavaScript	
Japan	Java	Python
Brazil	Python	JavaScript
Netherlands	Go	Python
USA	Ruby	JavaScript
Pakistan	C++	Python
Sweden	Python	Julia
India	JavaScript	HTML
South Africa	Python	JavaScript

```
sqlite>
```

Operation 03

```
SELECT *FROM "developers" WHERE "annual_salary" >
      '50000';
```

```
sqlite> SELECT * FROM "developers" WHERE "annual_salary" > '50000';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
6	UK	12	Python	R	PostgreSQL	Flask	85000
7	Canada	4	JavaScript	Python	MongoDB	Vue	60000
9	USA	15	C++	Python	SQLite	Qt	110000
11	France	8	Python	SQL	PostgreSQL	Django	65000
12	Australia	9	JavaScript		MongoDB	React	75000
13	Japan	11	Java	Python	Oracle	Spring	82000
15	Netherlands	6	Go	Python	PostgreSQL		68000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
18	Sweden	10	Python	Julia	SQLite	Django	90000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 04

```
SELECT *FROM "developers" WHERE "annual_salary" >
      '50000' AND "country" = 'Pakistan';
```

```
sqlite> SELECT * FROM "developers" WHERE "annual_salary" < '20000' AND "country" = 'Pakistan';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
5	Pakistan	1	C				3000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
17	Pakistan	5	C++	Python	MySQL		14000

```
sqlite>
```

Operation 05

```
SELECT * FROM "developers" WHERE
("primary_language" = 'Python' OR "secondary_language"
 = 'Python') AND "database_used" = 'PostgreSQL';
```

```
sqlite> SELECT * FROM "developers" WHERE ("primary_language" = 'Python' OR "secondary_language" = 'Python')
...> AND "database_used" = 'PostgreSQL';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
6	UK	12	Python	R	PostgreSQL	Flask	85000
11	France	8	Python	SQL	PostgreSQL	Django	65000
15	Netherlands	6	Go	Python	PostgreSQL		68000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 06

SELECT * FROM “developers” WHERE (“country” = ‘USA’ OR “country” = ‘Germany’) AND “annual_salary” > ‘60000’;

```
sqlite> SELECT * FROM "developers" WHERE ("country" = 'USA' OR "country" = 'Germany') AND "annual_salary" > '60000';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
9	USA	15	C++	Python	SQLite	Qt	110000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000

Operation 07

SELECT * FROM “developers” WHERE NOT
“primary_language” = ‘JavaScript’ AND NOT
“secondary_language” = ‘JavaScript’;

```
sqlite> SELECT * FROM "developers" WHERE NOT "primary_language" = 'JavaScript' AND NOT "secondary_language" = 'JavaScript';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
2	India	5	Java	SQL	Oracle	Spring	18000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
6	UK	12	Python	R	PostgreSQL	Flask	85000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
9	USA	15	C++	Python	SQLite	Qt	110000
11	France	8	Python	SQL	PostgreSQL	Django	65000
13	Japan	11	Java	Python	Oracle	Spring	82000
15	Netherlands	6	Go	Python	PostgreSQL		68000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000

```
sqlite>
```

Operation 08

SELECT * FROM “developers” WHERE
“experience_years” BETWEEN 5 AND 10;

```
sqlite> SELECT * FROM "developers" WHERE "experience_years" BETWEEN 5 AND 10;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
2	India	5	Java	SQL	Oracle	Spring	18000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
11	France	8	Python	SQL	PostgreSQL	Django	65000
12	Australia	9	JavaScript		MongoDB	React	75000
15	Netherlands	6	Go	Python	PostgreSQL		68000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 09

SELECT * FROM “developers” WHERE
“primary_language” IN (‘Python’, ‘C++’, ‘Java’, ‘PHP’);

```
sqlite> SELECT * FROM 'developers' WHERE "primary_language" IN ('Python', 'Java', 'C++', 'PHP');
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
2	India	5	Java	SQL	Oracle	Spring	18000
6	UK	12	Python	R	PostgreSQL	Flask	85000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
9	USA	15	C++	Python	SQLite	Qt	110000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
11	France	8	Python	SQL	PostgreSQL	Django	65000
13	Japan	11	Java	Python	Oracle	Spring	82000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 10

SELECT * FROM “developers” WHERE “framework” LIKE
‘S%’;

```
sqlite> SELECT * FROM 'developers' WHERE "framework" LIKE 'S%';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
2	India	5	Java	SQL	Oracle	Spring	18000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
13	Japan	11	Java	Python	Oracle	Spring	82000

```
sqlite>
```

Operation 11

SELECT * FROM “developers” WHERE “database_used”
LIKE ‘%Sql%’;

```
sqlite> SELECT * FROM 'developers' WHERE "database_used" LIKE '%Sql%';
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
6	UK	12	Python	R	PostgreSQL	Flask	85000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
9	USA	15	C++	Python	SQLite	Qt	110000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
11	France	8	Python	SQL	PostgreSQL	Django	65000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
15	Netherlands	6	Go	Python	PostgreSQL		68000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 12

SELECT * FROM “ developers” ORDER BY
‘annual_salary” DESC LIMIT 5;

```
sqlite> SELECT * FROM 'developers' ORDER BY "annual_salary" DESC LIMIT 5;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
9	USA	15	C++	Python	SQLite	Qt	110000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
18	Sweden	10	Python	Julia	SQLite	Django	90000
6	UK	12	Python	R	PostgreSQL	Flask	85000
13	Japan	11	Java	Python	Oracle	Spring	82000

```
sqlite>
```

Operation 13

SELECT * FROM “ developers” ORDER BY
‘annual_salary” ASC;

```
sqlite> SELECT * FROM 'developers' ORDER BY "annual_salary" ASC;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
5	Pakistan	1	C				3000
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
19	India	2	JavaScript	HTML	Firebase	Angular	10000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
17	Pakistan	5	C++	Python	MySQL		14000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
2	India	5	Java	SQL	Oracle	Spring	18000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000
7	Canada	4	JavaScript	Python	MongoDB	Vue	60000
11	France	8	Python	SQL	PostgreSQL	Django	65000
15	Netherlands	6	Go	Python	PostgreSQL		68000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
12	Australia	9	JavaScript		MongoDB	React	75000
13	Japan	11	Java	Python	Oracle	Spring	82000
6	UK	12	Python	R	PostgreSQL	Flask	85000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
18	Sweden	10	Python	Julia	SQLite	Django	90000
9	USA	15	C++	Python	SQLite	Qt	110000

```
sqlite>
```

Operation 14

SELECT * FROM “developers” ORDER BY
“experience_years” DESC LIMIT 1;

```
sqlite> SELECT * FROM 'developers' ORDER BY "experience_years" DESC LIMIT 1;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
9	USA	15	C++	Python	SQLite	Qt	110000

```
sqlite>
```

Operation 15

SELECT * FROM “developers” ORDER BY
“experience_years” DESC LIMIT 1 OFFSET 3;

```
sqlite> SELECT * FROM 'developers' ORDER BY "experience_years" DESC LIMIT 1 OFFSET 3;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000

```
sqlite>
```

Operation 16

SELECT * FROM “developers” WHERE “
primary_language” IS NOT NULL AND
“secondary_language” IS NOT NULL;

```
sqlite> SELECT * FROM 'developers' WHERE "primary_language" IS NOT NULL AND "secondary_language" IS NOT NULL;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
2	India	5	Java	SQL	Oracle	Spring	18000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
6	UK	12	Python	R	PostgreSQL	Flask	85000
7	Canada	4	JavaScript	Python	MongoDB	Vue	60000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
9	USA	15	C++	Python	SQLite	Qt	110000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
11	France	8	Python	SQL	PostgreSQL	Django	65000
13	Japan	11	Java	Python	Oracle	Spring	82000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
15	Netherlands	6	Go	Python	PostgreSQL		68000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
17	Pakistan	5	C++	Python	MySQL		14000
18	Sweden	10	Python	Julia	SQLite	Django	90000
19	India	2	JavaScript	HTML	Firebase	Angular	10000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000

```
sqlite>
```

Operation 17

SELECT * FROM “developers” WHERE “annual_salary”
IS NULL; (No such entry found)

```
sqlite> SELECT * FROM 'developers' WHERE "annual_salary" IS NULL;
```

```
sqlite> SELECT * FROM "developers" WHERE "annual_salary" IS NULL;
```

```
sqlite>
```

Operation 18

SELECT * FROM "developers" ORDER BY
"country" ASC, "annual_salary" DESC;

```
sqlite> SELECT * FROM "developers" ORDER BY "country" ASC, "annual_salary" DESC;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
12	Australia	9	JavaScript		MongoDB	React	75000
14	Brazil	4	Python	JavaScript	MySQL	Flask	22000
7	Canada	4	JavaScript	Python	MongoDB	Vue	60000
11	France	8	Python	SQL	PostgreSQL	Django	65000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
2	India	5	Java	SQL	Oracle	Spring	18000
10	India	6	PHP	JavaScript	MySQL	Laravel	15000
19	India	2	JavaScript	HTML	Firebase	Angular	10000
13	Japan	11	Java	Python	Oracle	Spring	82000
15	Netherlands	6	Go	Python	PostgreSQL		68000
17	Pakistan	5	C++	Python	MySQL		14000
8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000
1	Pakistan	2	Python	JavaScript	MySQL	Django	8000
5	Pakistan	1	C				3000
20	South Africa	7	Python	JavaScript	PostgreSQL	FastAPI	55000
18	Sweden	10	Python	Julia	SQLite	Django	90000
6	UK	12	Python	R	PostgreSQL	Flask	85000
9	USA	15	C++	Python	SQLite	Qt	110000
3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000

```
sqlite>
```

Operation 19

SELECT COUNT(DISTINCT "primary_language")
FROM "developers";

```
sqlite> SELECT count(DISTINCT "primary_language") FROM "developers";
```

count(DISTINCT "primary_language")
9

```
sqlite>
```

Operation 20

SELECT * FROM "developers" ORDER BY
"annual_salary" DESC LIMIT 5 OFFSET 5;

```
sqlite> SELECT * FROM "developers" ORDER BY "annual_salary" DESC LIMIT 5 OFFSET 5;
```

dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary
12	Australia	9	JavaScript		MongoDB	React	75000
16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000
4	Germany	7	C#	SQL	SQL Server	.NET	70000
15	Netherlands	6	Go	Python	PostgreSQL		68000
11	France	8	Python	SQL	PostgreSQL	Django	65000

```
sqlite>
```


Operation 21

SELECT AVG(annual_salary) FROM “developers”
WHERE “primary_language” = ‘Java’ AND
“database_used” = ‘Oracle’;

```
sqlite> SELECT AVG(annual_salary) FROM "developers" WHERE "primary_language" = 'Java' AND "database_used" = 'Oracle';
+-----+
| AVG(annual_salary) |
+-----+
| 50000.0             |
+-----+
sqlite>
```

Operation 22

ALTER TABLE “developers” ADD COLUMN “Bonus”;

```
sqlite> ALTER TABLE "developers"
...> ADD COLUMN "Bonus";
sqlite> .schema "devalopers";
Usage: .schema [--indent? [--nosys? ?LIKE-PATTERN?
sqlite> .schema "devalopers"
sqlite> .schema "developers"
CREATE TABLE developers (
  dev_id INTEGER PRIMARY KEY,
  country TEXT,
  experience_years INTEGER,
  primary_language TEXT,
  secondary_language TEXT,
  database_used TEXT,
  framework TEXT,
  annual_salary INTEGER
, "Bonus");
sqlite>
```

SELECT annual_salary + (1000 * experience_years) AS
“Bonus” FROM “developers”;

```
sqlite> SELECT annual_salary + (1000 * experience_years) AS "Bonus" FROM "developers";
+-----+
| Bonus |
+-----+
| 10000 |
| 23000 |
| 100000|
| 77000 |
| 4000  |
| 97000 |
| 64000 |
| 15000 |
| 125000|
| 21000 |
| 73000 |
| 84000 |
| 93000 |
| 26000 |
| 74000 |
| 75000 |
| 19000 |
| 100000|
| 12000 |
| 62000 |
+-----+
```

DB Browser for SQLite - B:\DB Lab\sqlite-tools-win-x64-3510200\Tech_Survey.db

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close

Database Structure Browse Data Edit Pragma Execute SQL

Table: developers Filter in any column

	dev_id	country	experience_years	primary_language	secondary_language	database_used	framework	annual_salary	Bonus
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	1	Pakistan	2	Python	JavaScript	MySQL	Django	8000	10000
2	2	India	5	Java	SQL	Oracle	Spring	18000	23000
3	3	USA	10	JavaScript	TypeScript	PostgreSQL	React	90000	100000
4	4	Germany	7	C#	SQL	SQL Server	.NET	70000	77000
5	5	Pakistan	1	C	NULL	NULL	NULL	3000	4000
6	6	UK	12	Python	R	PostgreSQL	Flask	85000	97000
7	7	Canada	4	JavaScript	Python	MongoDB	Vue	60000	64000
8	8	Pakistan	3	Java	Kotlin	MySQL	Spring	12000	15000
9	9	USA	15	C++	Python	SQLite	Qt	110000	125000
10	10	India	6	PHP	JavaScript	MySQL	Laravel	15000	21000
11	11	France	8	Python	SQL	PostgreSQL	Django	65000	73000
12	12	Australia	9	JavaScript	NULL	MongoDB	React	75000	84000
13	13	Japan	11	Java	Python	Oracle	Spring	82000	93000
14	14	Brazil	4	Python	JavaScript	MySQL	Flask	22000	26000
15	15	Netherlands	6	Go	Python	PostgreSQL	NULL	68000	74000
16	16	USA	3	Ruby	JavaScript	PostgreSQL	Rails	72000	75000
17	17	Pakistan	5	C++	Python	MySQL	NULL	14000	19000
18	18	Sweden	10	Python	Julia	SQLite	Django	90000	100000

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